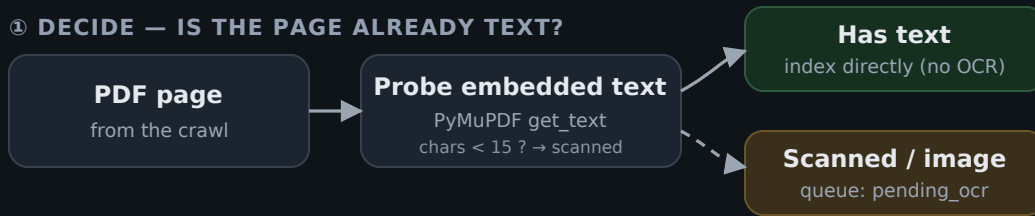


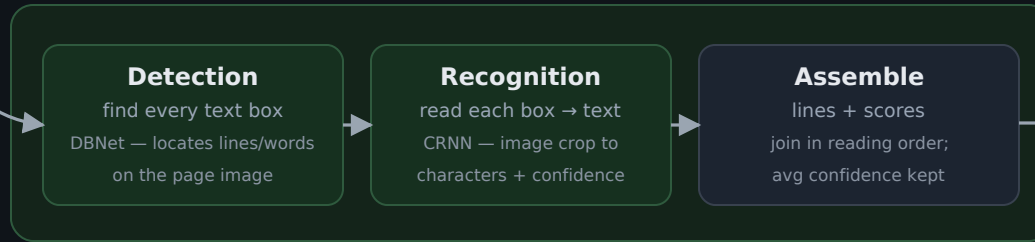
① DECIDE — IS THE PAGE ALREADY TEXT?



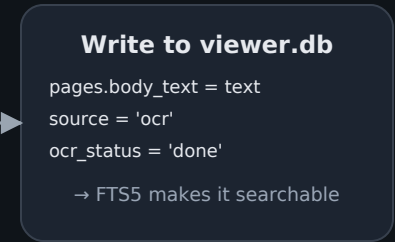
② RASTERIZE



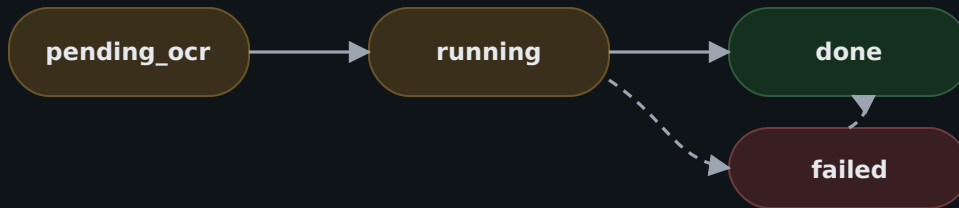
③ RapidOCR ENGINE (two neural stages, on CPU)



④ INDEX

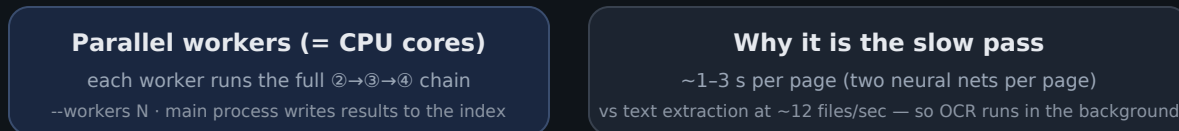


⑤ DURABLE STATE MACHINE (per page) — nothing lost if stopped



- committed every few pages → a crash/Ctrl+C just resumes
- re-running skips 'done' pages (idempotent)
- 'failed' pages keep last_error and are retryable

⑥ THROUGHPUT



queued / in-progress OCR compute / done render / workers index / page state failed (retryable)

Engine: RapidOCR (ONNX, pip-installed, no admin) — replaces Tesseract; Tesseract kept only as a fallback. All offline.

Verified on a scanned 815F parts page: 142 lines recovered at ~99% average confidence, including part numbers (5P8500, 9X2205).

Rule R1 — additive: no schema/data change, fallbacks preserved. Rule R2 — this is the detailed diagram for the OCR function.